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# THE SU(N) KOGUT–SUSSKIND VACUUM: CERTIFIED STRONG-COUPLING ENDPOINT AND SHORT-RANGE ENTANGLEMENT

*Main result: the strong-coupling threshold is reduced to a single certified constant $\kappa_Y$, $g_0(N) = (6N^2\kappa_Y/(N^2-1))^{1/4} \le (8\kappa_Y)^{1/4}$ for every N, with the link-embedding and boundary estimates proved; vacuum correlations cluster exponentially with an explicit correlation length, uniformly in volume; and an entanglement area law holds up to a polylogarithmic factor under an explicit energy-controlled locality hypothesis. Certifying $\kappa_Y$ and removing the logarithm remain open.*

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## Abstract

This paper is the strong-coupling companion to “Continuum 4D SU(N) Yang–Mills with a Mass Gap” [4]. It takes the four strong-coupling results of Provenance B8 of the weak-coupling programme [2] and the deliverable of workstream 1 of Provenance B2, a numerical threshold $g_0(N)$, and puts them on a single footing. Theorem 1 proves the link-embedding and boundary estimates required by Yarotsky's gap theorem and reduces $g_0(N)$ to one constant $\kappa_Y$ of that theorem, uniformly in N. Lemma 2 is a Lieb–Robinson bound with explicit constants despite unbounded electric terms. Theorem 3 proves exponential clustering of vacuum correlations, uniformly in volume, by adapting the Nachtergaele–Sims argument to unbounded on-site operators. Lemma 4 bounds the electric energy of every link by $O(g^{-4})$ and gives a volume-law entropy bound. Theorem 5 proves an entanglement area law up to a factor $(1 + \ln|A|)^3$ under an explicit energy-controlled locality hypothesis, and an exact area law under a further entangling-rate hypothesis. Definitions, statements and proofs are written in the mathematical English customary in Physical Review, combining the symbolic statements of the first version of this paper with the accompanying prose.


## Provenance

| Here | Origin | Status before this paper |
|---|---|---|
| Theorem 1: certified endpoint | B2, workstream 1; gate $\mathfrak{G}_1$ of [4, Lemma 1]; A1 of [2] | Open |
| Lemma 2: Lieb–Robinson bound | B8; [2, Lemma 5.1] | Proved, constants implicit |
| Theorem 3: exponential clustering | B8; [2, Corollary 5.2] | Tentative |
| Lemma 4: local electric energy | B8; [2, Lemma 5.3] | Proved |
| Theorem 5: entanglement area law | B8; [2, Proposition 5.4] | Tentative, three gaps |

Theorem 1 discharges the embedding part of gate $\mathfrak{G}_1$ in [4], used by Lemma 1(i), by the endpoint hypothesis Y of Lemma 4 and by Lemma 5(i) for $g \geq g_0$. The single remaining input is the numerical value of $\kappa_Y$.

# 0 Definitions

**Definition 1 (gauge group and normalization).** Fix $N \geq 2$ and let $G = SU(N)$, with the bi-invariant metric $\langle X, Y\rangle = -\mathrm{Tr}(XY)$ on $\mathfrak{su}(N)$. We write $D = \dim G = N^2 - 1$ and $c_N = (N^2 - 1)/N$, which is the smallest positive eigenvalue of $-\Delta_G$, and $\rho$ for the fundamental representation.

**Definition 2 (lattice).** For $L \in \mathbb{N}$ let $\Lambda_L = \{0, \dots, L\}^3$ be the open cubic box. Its positively oriented links are $E_L = \{(x, i) : x, x + e_i \in \Lambda_L\}$ and its plaquettes are $P_L = \{(x, i, j) : i < j, x + e_i + e_j \in \Lambda_L\}$. The oriented boundary of $p = (x, i, j)$ is $\partial p = ((x, i), (x + e_i, j), (x + e_j, i)^{-1}, (x, j)^{-1})$, and $U_p = \Pi_{e \in \partial p} U_e$ is the ordered plaquette holonomy.

**Definition 3 (Hilbert space and local terms).** The full Hilbert space is $\mathscr{H}_L = \otimes_{e \in E_L} L^2(G, dU)$, with normalized Haar measure $dU$. On each link we define the normalized electric operator $h_e = -c_N^{-1} \Delta_e$, whose ground state is the constant function and whose gap is exactly 1. The magnetic terms are $\psi_p = -(2/c_N)\,\mathrm{Re}\,\mathrm{Tr}\,\rho(U_p)$, we set $\lambda = g^{-4}$, and $\Omega_0 = \mathbb{1}$ denotes the electric (product) vacuum.

**Definition 4 (Hamiltonians).** The normalized Hamiltonian is $\bar{H}_L = \Sigma_e h_e + \lambda \Sigma_p \psi_p$. It is related to the Kogut–Susskind Hamiltonian by

$$H_L(g) = (g^2/2) \Sigma_e(-\Delta_e) + g^{-2} \Sigma_p[N - \mathrm{Re}\,\mathrm{Tr}\,\rho(U_p)] = (g^2 c_N/2)\, \bar{H}_L + g^{-2} N |P_L|\, \mathbb{1}.$$

**Definition 5 (spectral data).** $E_0 = \inf \mathrm{spec}\, \bar{H}_L$; $\Omega_L$ is a normalized ground state, $P_{0,L}$ its projection, $\gamma_L = \inf(\mathrm{spec}\, \bar{H}_L \setminus \{E_0\}) - E_0$ the normalized gap, and $\langle\cdot\rangle_L = \langle \Omega_L, \cdot\, \Omega_L\rangle$.

**Definition 6 (link-site embedding).** The map $\iota : E_L \to \mathbb{Z}^3$, $\iota(x, i) = 2x + e_i$, places each link at a site of the enlarged lattice. We set $\Lambda_0 = \{0, 1, 2\}^3$ and $B_L = \{0, \dots, 2L + 2\}^3$, and group the magnetic terms by lower corner: $\varphi_{2x} = \lambda \Sigma_{(x,i,j) \in P_L} \psi_{(x,i,j)}$, with $\varphi_z = 0$ for $z \notin 2\mathbb{Z}^3$.

**Definition 7 (Yarotsky class) [3, §1].** For $\beta > 0$, $\mathscr{Y}(\beta)$ is the set of Hamiltonians $\Sigma_{z \in \Lambda} h_z + \Sigma_{z + \Lambda_0 \subset \Lambda} \varphi_z$ on finite $\Lambda \subset \mathbb{Z}^3$ in which each $h_z$ is self-adjoint and nonnegative, with a nondegenerate kernel and $\mathrm{spec}\, h_z \cap (0, 1) = \varnothing$, and each $\varphi_z$ is a bounded self-adjoint operator supported in $z + \Lambda_0$ with $\sup_z \|\varphi_z\| \leq \beta$.

**Definition 8 (Yarotsky constants) [3, Thm 1].** Let $c_1, c_2 > 0$ be constants such that for every $\beta < c_1$, every finite $\Lambda$ and every $H \in \mathscr{Y}(\beta)$, the ground state of $H$ is nondegenerate and $\mathrm{gap}(H) \geq 1 - c_2\beta$. We define the single certification constant $\kappa_Y = \max\{2/c_1, 2c_2\}$ and $\beta^* = \kappa_Y^{-1} = \min\{c_1/2, (2c_2)^{-1}\}$.

**Definition 9 (threshold).** $g_0(N) = (6N\kappa_Y/c_N)^{1/4} = (6N^2\kappa_Y/(N^2 - 1))^{1/4}$.

**Definition 10 (locality).** On $\mathbb{Z}^3$ we use the metric $d(z, z') = \|z - z'\|_\infty$. The local algebra of $X \subset \mathbb{Z}^3$ is $\mathscr{A}_X = \mathscr{B}(\otimes_{z \in X} L^2(G)) \otimes \mathbb{1}$. For $\mu > 0$ the interaction norm is $\|\Phi\|_\mu = \sup_x \Sigma_{z\,:\,x \in z + \Lambda_0} \|\varphi_z\|\, e^{\mu\, \mathrm{diam}(z + \Lambda_0)}$. The Heisenberg dynamics are $\tau_t(A) = e^{it\bar{H}_L} A e^{-it\bar{H}_L}$ and $\tau^0_t(A) = e^{itH_0} A e^{-itH_0}$, with $H_0 = \Sigma_e h_e$.

**Definition 11 (energy and entropy functions).** $e_4(N, \lambda) = 16N\lambda/c_N + 256N(N^2 - 1)\lambda^2/c_N^3$. The single-link partition function is $Z(\beta) = \mathrm{Tr}_{L^2(G)}\, e^{-\beta h} = \Sigma_{\pi \in \hat{G}} d_\pi^2\, e^{-\beta C(\pi)/c_N}$, and $s(E) = \inf_{\beta > 0}[\beta E + \ln Z(\beta)]$. The heat-kernel asymptotics on $G$ [10] give a constant $c_Z$ with $\ln Z(\beta) \leq (D/2) \ln \beta^{-1} + c_Z$ for all $\beta \in (0, 1]$.

**Definition 12 (regions).** For $A \subset E_L$ with complement $A^c$, $\rho_A = \mathrm{Tr}_{A^c}|\Omega_L\rangle\langle\Omega_L|$ is the reduced vacuum state, $S(\rho) = -\mathrm{Tr}\,\rho \ln \rho$ the von Neumann entropy, and $h(\varepsilon) = -\varepsilon \ln \varepsilon - (1-\varepsilon)\ln(1-\varepsilon)$ the binary entropy. The boundary of $A$ is $\partial A = \{e \in A : d(\iota(e), \iota(e')) \le 2 \text{ for some } e' \in A^c\}$, and the strip of width $w$ is $S_w(A) = \{e \in E_L : d(\iota(e), \iota(\partial A)) \le w\}$.

**Definition 13 (energy-controlled locality, hypothesis $\mathfrak{U}_E$).** We say $\mathfrak{U}_E(\ell, C_{\mathfrak{U}}, E_{\mathfrak{U}})$ holds if for every $L$, every region $A$ and every $w \in \mathbb{N}$ there are unitaries $V_A \in \mathcal{A}_{\iota(A)}$, $V_B \in \mathcal{A}_{\iota(A^c)}$ and $W \in \mathcal{A}_{\iota(S_w(A))}$ such that the state $\sigma = (V_A \otimes V_B)W\Omega_0$ satisfies

$$\|\Omega_L - \sigma\| \le C_{\mathfrak{U}} |\partial A|\, e^{-w/\ell} \quad \text{and} \quad \sup_e \max\{\langle \sigma, h_e \sigma\rangle, \langle W\Omega_0, h_e W\Omega_0\rangle\} \le E_{\mathfrak{U}}.$$

**Definition 14 (entangling rate, hypothesis $\mathfrak{S}$).** Let $\tilde{H}(s) = H_0 + s\lambda \Sigma_p \psi_p$ for $s \in [0,1]$, with ground state $\Omega(s)$. We say $\mathfrak{S}(c_{\mathfrak{S}})$ holds if for every $L$ and $A$ the map $s \mapsto S(\rho_A(s))$ is absolutely continuous on $[0,1]$ and $|\partial_s S(\rho_A(s))| \le c_{\mathfrak{S}} |\partial A|$ almost everywhere.

# 1 Certified strong-coupling endpoint

**Theorem 1.**

(i) For every $g > 0$ and $L \in \mathbb{N}$, the embedding $\iota$ is injective, $\iota(\partial p) \subset 2x + \Lambda_0$ for every plaquette $p = (x, i, j)$, and $\sup_z \|\varphi_z\| \le 6N/(g^4 c_N)$, boundary plaquettes included.

(ii) If $g \ge g_0(N)$, then for every $L$ the ground state of $\bar{H}_L$ is nondegenerate and gauge invariant, $\gamma_L \ge \tfrac{1}{2}$, and $H_L(g) - E_{0,L} \ge (g^2 c_N/4)(\mathbb{1} - P_{0,L})$.

(iii) $g_0(2) = (8\kappa_Y)^{1/4}$ and $g_0(3) = (27\kappa_Y/4)^{1/4}$; $N \mapsto g_0(N)$ is decreasing, with $\lim_{N\to\infty} g_0(N) = (6\kappa_Y)^{1/4}$ and $\sup_{N\ge 2} g_0(N) = (8\kappa_Y)^{1/4}$.

**Proof.** *(i)* The k-th coordinate of $\iota(x, i) = 2x + e_i$ is odd exactly when $k = i$, so $\iota(x, i)$ determines both $i$ and $x$, and $\iota$ is injective. The four links of $p = (x, i, j)$ are sent to $2x + e_i$, $2x + 2e_i + e_j$, $2x + e_i + 2e_j$ and $2x + e_j$, all of which lie in $2x + \Lambda_0$. Since $\sup_{U\in G}|\mathrm{ReTr}\,\rho(U)| = N$, each magnetic term satisfies $\|\psi_p\| = 2N/c_N$, hence $\|\lambda\psi_p\| = 2N/(g^4 c_N)$. A lower corner $x$ carries at most the three plaquettes $(x, i, j)$ with $1 \le i < j \le 3$, and fewer at the boundary of $\Lambda_L$, so $\|\varphi_{2x}\| \le 6N/(g^4 c_N)$.

*(ii)* By Definition 9, $g \ge g_0(N)$ is equivalent to $6N/(g^4 c_N) \le \beta^*$. Moreover $\beta^* \le c_1/2 < c_1$ and $1 - c_2\beta^* \ge \tfrac{1}{2}$. Because $\iota(E_L) \subset \{0, \dots, 2L\}^3$, every nonzero $\varphi_z$ is supported in $z + \Lambda_0 \subset B_L$, so by (i) the enlarged Hamiltonian $H^{\mathrm{enl}} = \Sigma_{z\in B_L} h_z + \Sigma_z \varphi_z$ belongs to $\mathscr{Y}(\beta^*)$. Yarotsky's theorem [3, Thm 1] then gives a nondegenerate ground state and $\mathrm{gap}(H^{\mathrm{enl}}) \ge \tfrac{1}{2}$. With $R = B_L \setminus \iota(E_L)$, the enlarged Hamiltonian decouples exactly as $H^{\mathrm{enl}} = \bar{H}_L \otimes \mathbb{1} + \mathbb{1} \otimes \Sigma_{z\in R} h_z$. The second summand has a unique product ground state and gap 1, so $\mathrm{gap}(H^{\mathrm{enl}}) = \min\{\gamma_L, 1\}$, the ground state of $\bar{H}_L$ is nondegenerate, and $\gamma_L \ge \tfrac{1}{2}$. Since $H_L(g) = (g^2 c_N/2)\bar{H}_L + \mathrm{const}\cdot\mathbb{1}$, this gives $H_L(g) - E_{0,L} \ge (g^2 c_N/4)(\mathbb{1} - P_{0,L})$. Finally, $e^{-t\bar{H}_L}$ is positivity improving, so $\Omega_L$ can be chosen strictly positive. Every gauge transformation $\mathscr{U}(h)$ commutes with $\bar{H}_L$ and maps $\Omega_L$ to a positive normalized ground state, which by nondegeneracy is $\Omega_L$ itself. Hence $\Omega_L$ is physical, and the bound restricts to the physical subspace.

*(iii)* $6N/c_N = 6N^2/(N^2 - 1)$, which equals 8 for $N = 2$ and $27/4$ for $N = 3$. Since $\partial_N[N^2/(N^2 - 1)] = -2N/(N^2 - 1)^2 < 0$ and $N^2/(N^2 - 1) \to 1$, $g_0(N)$ decreases to $(6\kappa_Y)^{1/4}$, and its supremum is attained at $N = 2$. ∎

## 2 Lieb–Robinson bound

**Lemma 2.** For every $\mu > 0$, all finite $X, Y \subset \mathbb{Z}^3$, all $A \in \mathcal{A}_X$, $B \in \mathcal{A}_Y$ and all $t \in \mathbb{R}$,

$$\|[\tau_t(A), B]\| \le 2\|A\|\|B\| \min\{ \Sigma_{x\in X} e^{-\mu\, d(x,Y)}, \Sigma_{y\in Y} e^{-\mu\, d(X,y)} \} (e^{2\|\Phi\|_\mu |t|} - 1),$$

and the interaction norm satisfies $\|\Phi\|_\mu \le 8e^{2\mu} \cdot 6N/(g^4 c_N)$.

**Proof.** The magnetic term $\lambda\Sigma\psi_p$ is bounded and $H_0$ is self-adjoint, so by the Kato–Rellich theorem $\tilde{H}_L$ is self-adjoint on $\mathcal{D}(H_0)$. Let $W(t) = e^{-itH_0} e^{it\tilde{H}_L}$ and $\alpha_t(A) = W(t)AW(t)^*$. Then $\tau_t = \tau^0_t \circ \alpha_t$, and $W$ solves $\partial_t W(t) = i\, \Sigma_z\, \tau^0_{-t}(\varphi_z)\, W(t)$. The free dynamics $\tau^0_t = \otimes_z \mathrm{Ad}\, e^{ith_z}$ acts site by site, so it maps each $\mathcal{A}_X$ onto itself and preserves norms. The time-dependent interaction $\Phi(t) = \{\tau^0_{-t}(\varphi_z)\}$ therefore has the same supports as $\Phi$ and $\|\Phi(t)\|_\mu = \|\Phi\|_\mu$ for all $t$. The Lieb–Robinson bound for bounded time-dependent interactions [6] then gives $\|[\alpha_t(A), B']\| \le 2\|A\|\|B'\|\, \Sigma_{x\in X} e^{-\mu\, d(x,Y)} (e^{2\|\Phi\|_\mu |t|} - 1)$ for every $B' \in \mathcal{A}_Y$. Since $[\tau_t(A), B] = \tau^0_t([\alpha_t(A), \tau^0_{-t}(B)])$ with $\tau^0_{-t}(B) \in \mathcal{A}_Y$ and $\|\tau^0_{-t}(B)\| = \|B\|$, the bound with the sum over $X$ follows. The identity $[\tau_t(A), B] = -\tau_t([\tau_{-t}(B), A])$ gives the bound with the sum over $Y$.

For the norm estimate, a site $x$ lies in $z + \Lambda_0$ with $z \in 2\mathbb{Z}^3$ only if each coordinate $z_k$ belongs to $\{x_k - 2, x_k - 1, x_k\} \cap 2\mathbb{Z}$, which has at most two elements; hence at most $2^3 = 8$ grouped terms contain $x$. With $\mathrm{diam}_\infty\, \Lambda_0 = 2$ and Theorem 1(i), $\|\Phi\|_\mu \le 8e^{2\mu} \cdot 6N/(g^4 c_N)$. ∎

## 3 Exponential clustering

**Theorem 3.** Let $g \geq g_0(N)$, $\mu > 0$ and $\mu_3 = (\tfrac{1}{2}\mu)/(4\|\Phi\|_\mu + \tfrac{1}{2})$. Then for every $L$, all disjoint finite $X$, $Y$ and all $A \in \mathscr{A}_X$, $B \in \mathscr{A}_Y$,

$$|\langle AB\rangle_L - \langle A\rangle_L\langle B\rangle_L| \leq \|A\|\|B\|\,(1 + (2/\pi)\min\{|X|, |Y|\} + (\pi\mu_3\, d(X, Y))^{-1/2})\, e^{-\mu_3\, d(X,Y)}.$$

In particular, for $\mu = \tfrac{1}{2}$ the correlation length $\xi = \mu_3^{-1}$ obeys $\xi \leq 4(192eN/(g^4 c_N) + \tfrac{1}{2})$, uniformly in $L$.

**Proof.** We follow the exponential clustering theorem of Nachtergaele and Sims [5, Thm 2], whose proof uses only four ingredients: a Lieb–Robinson bound, a spectral gap above a nondegenerate ground state, entire analytic approximations of local observables, and the spectral theorem together with contour integration and dominated convergence [5, Lemma 1 and proof of Thm 2]. We verify each ingredient for $\bar{H}_L$.

First, Lemma 2 gives $\|[\tau_t(A), B]\| \leq 2\|A\|\|B\| \min\{|X|, |Y|\}\, e^{-\mu\, d(X,Y)}\, e^{2\|\Phi\|_\mu |t|}$, which has the form used in [5] with the decay rate $\lambda$ replaced by $\mu$ and the volume factor $|Y|$ replaced by $\min\{|X|, |Y|\}$. Second, by Theorem 1(ii), $H = \bar{H}_L - E_0 \geq 0$, $\ker H = \mathbb{C}\Omega_L$, and $\operatorname{spec} H \subset \{0\} \cup [\gamma, \infty)$ with $\gamma = \tfrac{1}{2}$. Replacing $B$ by $\tilde{B} = B - \langle B\rangle_L \mathbb{1}$ leaves the commutator unchanged and gives $P_{0,L}\tilde{B}\Omega_L = P_{0,L}\tilde{B}^*\Omega_L = 0$, as required in [5], while $\langle AB\rangle_L - \langle A\rangle_L\langle B\rangle_L = \langle \Omega_L, A\tilde{B}\Omega_L\rangle$.

Third, for $a > 0$ the Gaussian time average $\tilde{B}(a) = (a/\pi)^{1/2}\int_{\mathbb{R}} \tau_s(\tilde{B})e^{-as^2} ds$ extends to the entire function $z \mapsto \tau_z(\tilde{B}(a)) = (a/\pi)^{1/2}\int_{\mathbb{R}} \tau_s(\tilde{B})e^{-a(s-z)^2} ds$, with $\|\tau_z(\tilde{B}(a))\| \leq \|\tilde{B}\|e^{a(\operatorname{Im} z)^2}$. This construction uses only the real-time automorphisms $\tau_s$, $s \in \mathbb{R}$, so it is valid for any self-adjoint $\bar{H}_L$, bounded or not; neither boundedness of the Hamiltonian nor finite local dimension enters. This is the point at which the finite-dimensional setting of [5] is replaced. Fourth, the spectral theorem, contour integration and dominated convergence apply to any nonnegative self-adjoint $H$.

All ingredients of [5, Thm 2] therefore hold for $\bar{H}_L$, and its conclusion at imaginary time $b = 0$, with $\lambda \mapsto \mu$, $\|\Phi\|_\lambda \mapsto \|\Phi\|_\mu$, $\gamma \mapsto \tfrac{1}{2}$ and $|Y| \mapsto \min\{|X|, |Y|\}$, is the stated bound with $\mu_3 = (\tfrac{1}{2}\mu)/(4\|\Phi\|_\mu + \tfrac{1}{2})$. For $\mu = \tfrac{1}{2}$, Lemma 2 gives $4\|\Phi\|_{1/2} \leq 4 \cdot 8e \cdot 6N/(g^4 c_N) = 192eN/(g^4 c_N)$, hence $\mu_3 \geq \tfrac{1}{4}/(192eN/(g^4 c_N) + \tfrac{1}{2})$. ∎

## 4 Local electric energy

**Lemma 4.** (i) If $g \geq g_0(N)$, then $\langle h_e \rangle_L \leq e_4(N, \lambda)$ for every $L$ and every link $e \in E_L$.

(ii) For any state $\rho$ with $\mathrm{Tr}\,\rho(\Sigma_{e\in A} h_e) \leq E$, $S(\mathrm{Tr}_{A^c}\rho) \leq \inf_{\beta>0}[\beta E + |A| \ln Z(\beta)]$. Consequently $S(\rho_A) \leq |A|\, s(e_4)$.

**Proof.** *(i)* Fix a link $e$ and set $\bar{H}_{\setminus e} = \bar{H}_L - h_e$. Since $\Omega_L$ is normalized, $\langle h_e \rangle_L = E_0 - \langle \Omega_L, \bar{H}_{\setminus e}\Omega_L\rangle \leq E_0(\bar{H}_L) - E_0(\bar{H}_{\setminus e})$. The operator $\bar{H}_{\setminus e}$ commutes with multiplication by every continuous function of $U_e$, so it decomposes as a direct integral $\bar{H}_{\setminus e} = \int_G^{\oplus} K(u)du$ of operators $K(u)$ on the remaining links, with $U_e$ frozen at $u$; consequently $E_0(\bar{H}_{\setminus e}) = \min_{u\in G} E_0(K(u))$. Applying Theorem 1(i)–(ii) to $E_L \setminus \{e\}$ shows $K(u) \in \mathscr{Y}(\beta^*)$, so each $K(u)$ has a unique positive ground state $\varphi(u)$ and gap at least ½, uniformly in $u$. By analytic perturbation theory [11], $\varphi(u)$ is smooth in $u$.

The normalized trial state $\psi(U_e, U') = \varphi(U_e)(U')$ gives

$$E_0(\bar{H}_L) \leq \langle \psi, \bar{H}_L \psi\rangle = \int_G E_0(K(u))du + c_N^{-1}\int_G \|\nabla_u \varphi(u)\|^2 du \leq \max_u E_0(K(u)) + c_N^{-1} \sup_u \|\nabla_u \varphi\|^2.$$

Only the four plaquettes containing $e$ depend on $u$, and each $\lambda\psi_p$ varies over a range of width $4N\lambda/c_N$; hence $\|K(u) - K(u')\| \leq 16N\lambda/c_N$ and $\max_u E_0(K(u)) - \min_u E_0(K(u)) \leq 16N\lambda/c_N$. For a unit tangent vector $X$, first-order perturbation theory gives $\partial_X \varphi = -R(u)(\partial_X K)\varphi$ with $\|R(u)\| \leq 2$, and $|\mathrm{Re}\,\mathrm{Tr}(XW)| \leq \|X\|_1 \leq \sqrt{N}|X|$ gives $\|\partial_X(\lambda\psi_p)\| \leq 2\sqrt{N}\lambda/c_N$. Summing over the four plaquettes, $\|\partial_X \varphi\| \leq 16\sqrt{N}\lambda/c_N$, and summing over an orthonormal basis of dimension $N^2 - 1$, $\|\nabla_u \varphi\|^2 \leq 256N(N^2-1)\lambda^2/c_N^2$. Combining these estimates yields $\langle h_e \rangle_L \leq e_4(N, \lambda)$.

*(ii)* Let $\sigma_\beta = e^{-\beta\Sigma_{e\in A} h_e}/Z(\beta)^{|A|}$, a density matrix for every $\beta > 0$. Nonnegativity of the relative entropy, which holds in infinite dimensions [9], gives $S(\rho_A) \leq -\mathrm{Tr}\,\rho_A \ln \sigma_\beta = \beta\,\mathrm{Tr}(\rho_A \Sigma h_e) + |A| \ln Z(\beta)$. For the vacuum, part (i) gives $\mathrm{Tr}(\rho_A \Sigma_{e\in A} h_e) \leq |A| e_4$. ∎

## 5 Entanglement area law

**Theorem 5.** (i) Let $g \geq g_0(N)$ and assume $\mathfrak{A}_E(\ell, C_{\mathfrak{A}}, E_{\mathfrak{A}})$. Then for every L and every region $A \subset E_L$ with $|A| \geq 2$ and $|\partial A| \geq 1$,

$$S(\rho_A) \leq (2\lceil \ell \ln|A| \rceil + 1)^3 |\partial A|\, s(E_{\mathfrak{A}}) + C_{\mathfrak{A}}|\partial A|(D \ln|A| + 2c_Z) + 2\max\{e_4, E_{\mathfrak{A}}\} + \ln 2 + C_{\mathfrak{A}}|\partial A|\, s(e_4).$$

In particular, there is $c_5 = c_5(N, \ell, C_{\mathfrak{A}}, E_{\mathfrak{A}})$ with $S(\rho_A) \leq c_5|\partial A|(1 + \ln|A|)^3$.

(ii) If $\Xi(c_\Xi)$ holds, then $S(\rho_A) \leq c_\Xi|\partial A|$ for every L and A.

**Proof.** *(i)* Choose $w = \lceil \ell \ln|A| \rceil$ and let $\varepsilon = C_{\mathfrak{A}}|\partial A| e^{-w/\ell} \leq C_{\mathfrak{A}}|\partial A|/|A|$. If $\varepsilon > 1$, then $|A| < C_{\mathfrak{A}}|\partial A|$, and Lemma 4(ii) gives $S(\rho_A) \leq |A| s(e_4) < C_{\mathfrak{A}}|\partial A| s(e_4)$, which is covered by the last term of the bound. We may therefore assume $\varepsilon \leq 1$.

Let $\sigma = (V_A \otimes V_B) W \Omega_0$ be the state supplied by $\mathfrak{A}_E$, and $\sigma_A$ its reduced state on A. Since $V_A \otimes V_B$ is a product of unitaries acting on A and on $A^c$, $\sigma_A$ is unitarily equivalent to the reduced state of $W\Omega_0$, so $S(\sigma_A) = S(\mathrm{Tr}_{A^c} |W\Omega_0\rangle\langle W\Omega_0|)$. The electric vacuum is a product state and W acts only on the strip, so $W\Omega_0$ factorizes into a pure state on $S_w(A)$ tensored with the vacuum elsewhere. The entropy of $\sigma_A$ is therefore the entropy of a reduced state on $S_w(A) \cap A$, whose electric energy per link is at most $E_{\mathfrak{A}}$ by $\mathfrak{A}_E$. The Gibbs bound of Lemma 4(ii) gives $S(\sigma_A) \leq |S_w(A) \cap A|\, s(E_{\mathfrak{A}})$, and since a d-ball of radius w in $\mathbb{Z}^3$ contains $(2w + 1)^3$ sites, $|S_w(A)| \leq (2w + 1)^3 |\partial A|$.

Next we compare $\rho_A$ with $\sigma_A$. Partial trace does not increase trace distance, and for pure states $\tfrac{1}{2}\| |\Omega_L\rangle\langle\Omega_L| - |\sigma\rangle\langle\sigma| \|_1 \leq \|\Omega_L - \sigma\|$, so $\tfrac{1}{2}\|\rho_A - \sigma_A\|_1 \leq \varepsilon$. By Lemma 4(i) and $\mathfrak{A}_E$, both states have electric energy on A at most $\mathscr{E} = |A| \max\{e_4, E_{\mathfrak{A}}\}$. Winter's energy-constrained continuity bound [8, Lemma 15] gives $|S(\rho_A) - S(\sigma_A)| \leq 2\varepsilon S(\gamma(\mathscr{E}/\varepsilon)) + h(\varepsilon)$, where $\gamma(E')$ is the Gibbs state of energy $E'$ and $S(\gamma(E')) \leq \beta E' + |A| \ln Z(\beta)$ for every $\beta > 0$. Taking $\beta = |A|^{-1} \leq 1$ and using Definition 11,

$$2\varepsilon S(\gamma(\mathscr{E}/\varepsilon)) \leq 2\mathscr{E}/|A| + 2\varepsilon|A|((D/2)\ln|A| + c_Z) \leq 2\max\{e_4, E_{\mathfrak{A}}\} + C_{\mathfrak{A}}|\partial A|(D \ln|A| + 2c_Z),$$

and $h(\varepsilon) \leq \ln 2$. The triangle inequality $S(\rho_A) \leq S(\sigma_A) + |S(\rho_A) - S(\sigma_A)|$ then gives the stated bound. Since $w \leq \ell \ln|A| + 1$, $(2w + 1)^3 \leq (2\ell + 3)^3 (1 + \ln|A|)^3$, which yields the constant $c_5$.

*(ii)* At $s = 0$ the Hamiltonian is $H_0$, whose ground state $\Omega_0$ is a product state, so $S(\rho_A(0)) = 0$. At $s = 1$, $\Omega(1) = \Omega_L$. By $\Xi$, $S(\rho_A) \leq \int_0^1 |\partial_s S(\rho_A(s))| ds \leq c_\Xi |\partial A|$. ∎

## 6 Status

| Result | Status | Remaining input |
|---|---|---|
| Theorem 1: certified endpoint | Proved | Numerical value of the single constant $\kappa_Y = \max\{2/c_1, 2c_2\}$ from the proof of [3, Thm 1]. The embedding, boundary and N-uniformity estimates are complete. |
| Lemma 2: Lieb–Robinson bound | Proved | Uses the time-dependent Lieb–Robinson theorem of [6] for bounded interactions. |
| Theorem 3: exponential clustering | Proved by adaptation | The argument of [5, Thm 2] is used with Lemma 2 in place of its Lieb–Robinson bound and with Gaussian time-smoothing in place of analyticity of bounded dynamics. [5] is stated for finite-dimensional sites; a line-by-line check of the adapted ingredients (third and fourth paragraphs of the proof) against [5] is recommended before publication. |
| Lemma 4: local electric energy | Proved | None beyond Theorem 1. |
| Theorem 5(i): area law up to $(1 + \ln\|A\|)^3$ | Conditional | Hypothesis $\mathfrak{A}_E$: a quasi-local decomposition of the vacuum with energy control. Automorphic equivalence [7] supplies the decomposition; the energy bounds on $\sigma$ and $W\Omega_0$ are open. |
| Theorem 5(ii): exact area law | Conditional | Hypothesis $\mathfrak{S}$: an energy-constrained small-incremental-entangling bound. Known proofs use finite local dimension. |

Compared with Proposition 5.4 of [2], which claimed $S(\rho_A) \le C|\partial A|$ with three open gaps, Theorem 5 makes the cost of infinite-dimensional links explicit: entropy continuity in infinite dimensions requires energy control, and the continuity error, of order $\varepsilon|A| \ln|A|$, forces a strip of width $\ln|A|$. Removing the polylogarithm is equivalent in difficulty to $\mathfrak{S}$.

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## Repository, citation and licence

GitHub repository: https://github.com/teknoquants/THE-SU-N-KOGUT-SUSSKIND-VACUUM-CERTIFIED-STRONG-COUPLING-ENDPOINT-AND-SHORT-RANGE-ENTANGLEMENT
DOI: https://doi.org/10.5281/zenodo.24115051
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Validation scripts: the scripts of the first version apply unchanged and are maintained in its repository: https://github.com/teknoquants/CERTIFIED-STRONG-COUPLING-ENDPOINT-AND-SHORT-RANGE-ENTANGLEMENT-FOR-THE-SU-N-KOGUT-SUSSKIND-VACUUM
(first version: https://doi.org/10.5281/zenodo.24105169).

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